

1. A sample consists of
 - (a) All units of the population
 - (b) 5% units of the population
 - (c) 10% units of the population
 - (d) Any fraction of the population**
2. Sampling is used in the situations
 - (a) Blood test of the patients
 - (b) Cooking rice in an utensil
 - (c) Purchase of food commodity from shopkeeper
 - (d) All the above**
3. The number of possible samples of size n out of N population size in SRSWOR is equal to
 - (a) $N C_n$** **answer is (a)**
 - (b) N^n
 - (c) $(N-n)/N$
 - (d) n/N
4. The number of possible samples of size n out of N population size in SRSWR is equal to
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(d) n/N

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(a) Nc_n

(b) N^n

(c) $(N-n)/N$

(e) n/N

5. The number of possible samples of size 2 out of 5 population size in SRSWOR is equal to

(a) 10

(b) 4

(c) 2

6. The number of possible samples of size 2 out of 5 population size in SRSWR is equal to
- (a) 25
 - (b) 20
 - (c) 2
 - (d) 12
7. Probability of a drawing unit at each subsequent draw remains same in
- (a) SRSWOR
 - (b) SRSWR
 - (c) Both (a) & (b)
 - (d) None
8. The sampling fraction in usual notation is expressed as
- (a) n/N
 - (b) N/n
 - (c) $1-n/N$
 - (d) None.
9. The finite population correction in usual notation is expressed as
- (a) $(N-n)/N$
 - (b) $1-(n/N)$
 - (c) Both (a) & (b)
 - (d) None

(d) None

10. A selection procedure of sampling having no involvement of probability is known as

- (a) SRSWOR
- (b) Purposive sampling**
- (c) SRSWR
- (d) None

11. For gathering information on rare events, sampling is used

- (a) SRSWOR
- (b) Stratified random sampling
- (c) Inverse sampling**
- (d) None

12. If a larger units have more probability of their inclusion in the sample, the sampling is known as

- (a) SRSWOR
- (b) PPS sampling**

- (d) None
13. Simple random samples can be drawn with the help of
- (a) Random numbers table
 - (b) Chit Method
 - (c) Roulette wheel
 - (d) All the above**
14. Sampling frame is a list of
- (a) A list of units of a population**
 - (b) A list of random numbers
 - (c) A list of natural numbers
 - (d) None
15. In SRSWR, the same sampling unit may be included in the sample
- (a) Only once
 - (b) Two times
 - (c) More than once**
 - (d) None
16. The discrepancies between the estimate and the population parameter is known as
- (a) Sampling error**
 - (b) Non-sampling error
 - (c) Formula error
 - (d) None
17. The error in a survey other than sampling error is known as
- (a) Sampling error
 - (b) Non-sampling error**

(c) Sampling error

(b) Non-sampling error

(c) Formula error

(d) None

18. A function of sample observations is known as

(a) Statistic

(b) Estimator

(c) Both (a)&(b)

(d) None

19. If the sample sizes are large from the population, then which error will contribute more errors

(a) Sampling error

(b) Non-sampling error

(c) Both(a)&(b)

(d) None

20. If the sample sizes are large from the population, then which error will contribute less errors
- (a) **Sampling error**
 - (b) Non-sampling error
 - (c) Both (a)&(b)
 - (d) None
21. Simple random sample can be drawn with the help of (a) random number tables (b) Chit method (c) roulette wheel (d) **all the above**
22. If each and every unit of a population has an equal chance of being included in the sample, it is called (a) restricted sampling (b) **unrestricted sampling** (c) purposive sampling (d) subjective sampling
23. If the observations recorded on five sampled items are 3, 4, 5, 6, 7, then sample variance is equal to (a) 0 (b) 1 (c) 2 (d) **2.5**
24. If all the observations in a set of observations are the same, then variance of set of values is (a) **0** (b) 2 (c) infinite (d) none
25. If the sample values are 1, 3, 5, 7, 9 then S.E. of sample mean is (a) 2 (b) $\sqrt{2}$ (c) 3 (d) $\sqrt{3}$ *the answer is (b)*
26. As a normal practice sampling fraction is considered to be negligible if it is (a) more than 5% (b) **less than 5%** (c) More than 10% (d) none
27. Systematic sampling is used when (a) when data are on cards (b) when the items are in row (c) when the items are given in a sequential order (d) **all the above**

above

28. If the population units N is multiple of n and k , then we use (a) **linear systematic sampling** (b) circular systematic sampling (c) random systematic sampling (d) all the above
29. Circular systematic sampling is used when (a) N is a whole number (b) **N is not divisible by n** (c) N is a multiple of n (d) All the above.
30. Problem of non-response has (a) no solution (b) **can be solved** (c) no meaning (d) none
31. If sample sizes increase, then sampling error will (a) increase (b) **decrease** (c) both (a) & (b) (d) none
32. If sample sizes increase, then non-sampling error will (a) **increase** (b) decrease (c) both (a) & (b) (d) none
33. A population is divided into clusters and it has been found that all the units within a cluster are same. In this situation which sampling will be adopted (a) SRSWOR (b) **Stratified random sampling** (c) Cluster sampling (d) Systematic sampling
34. A population N is divided into k strata. A sample of size n is to be chosen and N_i is the size of the i th stratum. Then sample size n as per proportional allocation is given by (a) $n_i = nN$ (b) **$n_i/N_i = n/N$** (c) $n_i N_i = nN$ (d) none

35. In case of iniverse sampling, the proportion p of m units of interest contained in a sample of n units is equal to (a) m/n (b) $(m-1)/n$ (c) **$(m-1)/(n-1)$** (d) $(m-1)/(n+1)$
36. If the respondents do not provide the required information to the researcher, then it is known as (a) non-sampling error (b) **the problem of non-response** (c) both (a) & (b) (d) none
37. The errors falling under faulty planning of survey, it is called (a) **non-sampling errors** (b) non-response errors (c) Sampling errors (d) Absolute error
38. If there is a certain number of very high values in the sample, it is preferable to compute (a) **Standard error** (b) Standard deviation (c) variance (d) all the above.
39. For estimating the population mean T , let T_1 be the sample mean under SRSWOR and T_2 sample mean under SRSWR, then which relationship is true (a) $\text{Var}(T_1) < \text{Var}(T_2)$ (b) $\text{Var}(T_1) > \text{Var}(T_2)$ (c) **$\text{Var}(T_1) = \text{Var}(T_2)$** (d) none
40. The magnitude of the standard error of an estimate is an index of its (a) accuracy (b) **precision** (c) efficiency (d) none
41. Which of the following statement is true (a) population mean increases with increase in sample size (b) population mean decreases with increase in sample size (c) population mean decreases with decrease in sample size (d) **population mean is a constant value.**
42. A sample of 25 units from a population with standard deviation 10 results into a total score of 450. Then the mean of sampling distribution is equal to (a) 45 (b) **18** (c) 50 (d) none
43. A population is perfectly homogeneous with respect to a characteristic, what size of sample would you need (a) no sample (b) a large sample (c) a small sample

- total score of 450. Then the mean of sampling distribution is equal to (a) 45(b) **18**
(c) 50 (d) none
43. A population is perfectly homogeneous with respect to a characteristic, what size of sample would you need (a) no sample (b) a large sample (c) a small sample
(d) **a single sample.**

SOME MORE QUESTIONS

44. A sample constant representing a population parameter is called **estimate.**
45. An estimator T_n which is most concentrated to the parameter θ is called **best estimator**
46. If we have purposive sample, estimation is **not possible.**
47. Estimation is possible only in case of **random sample.**
48. An estimator itself a **random variable.**
49. A value of an estimator is called an **estimate.**
50. If the value of an estimator T_n is equal to the parameter θ , then T_n is called **unbiased estimator** of the parameter θ .
51. Sampling is useful in **many** situations.
52. In a hypothetical population, all sampling units are **imagined.**
53. If the number of units in a population is limited, it is called **finite population.**

54. If the number of units in a population is unlimited, it is called **infinite population**.
55. If all the units of a population are surveyed, it is called **complete enumeration**.
56. The errors other than sampling errors is called **non sampling errors**.
57. The discrepancy between parameter and its estimate due to sampling process is known as **sampling errors**.
58. Any population constants is known as **parameter**.
59. A function based on sample values for estimating a parameter is called an **estimator**.
60. The specific value of an estimator is called an **estimate**.
61. Standard deviation of all possible estimates from sample of fixed size is called **standard error**.
62. Standard error of sample mean based on n sizes with standard deviation s is known as $s/\sqrt{n}$.
63. The list of all items of a population is known as **sampling frame**.
64. Another name of population is **universe**.
65. Number of possible samples of size n out of N using SRSWOR is $N C n$.
66. Probability of selection of any one sample out of $N C n$ is equal to $1/N C n$.
67. Number of possible samples of size n out of N using SRSWR is N^n .
68. Under SRSWOR, the same item can occur **more than once**.
69. The sampling procedure in which the population is divided into homogeneous groups and sample drawn from each group is called **stratified random**.

sampling.

- 70. Stratified random sampling is useful when population is **heterogeneous**.
- 71. Stratification is done with respect to certain **characteristics**.
- 72. Deciding the sample size for each stratum is known as **allocation problem**.
- 73. If the sample size of each stratum is in proportion to stratum size, it is called **proportional allocation**.
- 74. Stratified random sampling falls under the category of **restricted sampling**.
- 75. More heterogeneous is the population, larger **sample sizes** are required.
- 76. In usual notation $(N-n)/N$ is known as **finite population correction**.
- 77. In usual notation n/N is called **sampling fraction**.
- 78. For a high precision of estimates, **larger samples** are required.
- 79. Estimators and estimates are **different**.
- 80. Determination of sample sizes for each stratum subject to the cost constrained is called **optimum allocation**.
- 81. Optimum allocation is also known as **Neyman allocation**.
- 82. Double sampling is termed as **two phase sampling**.
- 83. Cluster sampling ordinarily leads to the loss of **precision**.
- 84. Cluster sampling helps to reduce **cost of survey**.

Mathematician

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85. Larger the cluster size, **less efficient** it is relative to the element as sampling unit.
86. Two stage sampling is less efficient rather than that of **single stage sampling**.
87. A sampling procedure in which units are selected with chance of selection in proportion to some measure of their size is known as **PPS sampling**.

MCOs. STA632

1). How many major types of Sampling?

1. 4

2. 2 ✓

3. 8

4. 3

2). Systematic Sampling is a type of

1. Probability Sampling ✓

2. Non-Probability Sampling.

3). A sample is a representative part of —

1. Population ✓

2. Self-distribution

4). Quota Sampling is —

15:17

2). Self-distribution

4) Quota Sampling is a type of

1. Probability
2. Non-Probability ✓

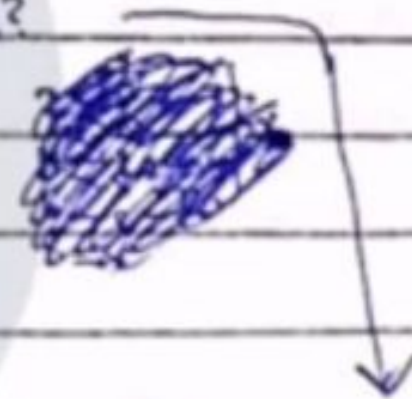
5) A numerical quantity computed from population is:

- | | |
|----------------|--------------|
| 1. Parameter ✓ | 3. Statistic |
| 2. Observation | 4. Variable |

6). If we have a population 2, 4, 6, 8 and $n=2$.
How many possible samples will be selected

6). If we have a population 2, 4, 6, 8 and $n=2$.
How many possible samples will be selected
from sampling with replacement?

- 1. 12
- 2. 8
- 3. 16 ✓
- 4. 6



7). Cluster Sampling is a type of

- 1. Probability Sampling ✓
- 2. Non-Probability Sampling



8) In Sampling with replacement a sampling unit can be selected:

1. Only once
2. less than one
3. more than one time ✓
4. none of these.

9) Which one of these sampling is probability method

1. Quota Sampling
2. Convenience sampling
3. Cluster Sampling ✓
4. Judgment sampling

10) The other name of standard deviation of a sampling distribution is

1. Replication Error
2. Sampling Error
3. Standard Error ✓
4. Sample Standard Deviation

11) If we have population 1, 2, 3 and $n=2$. How many samples will be selected from sampling without replacement.

1. 3 ✓
2. 6

3. 9
4. 8

$$\begin{array}{l} 1 \leftarrow 2 \\ 2 \leftarrow 3 \\ 3 \leftarrow x = 3 \end{array}$$

12). For the following population $x \leftarrow c(1, 2, 3, 4, 5)$, what will be R command for finding mean?

1. `mean(c)`

2. `mean(x)` ✓

3. `mean x`

4. `mean{x}`

13) If $\text{Var}(\bar{y}_{\text{asy}}) < \text{Var}(\bar{y}_{\text{hrs}})$

13) If $\text{Var}(\bar{y}_{cy}) < \text{Var}(\bar{y}_{or})$

1. $p_w \geq \frac{1}{(nk-1)}$

2. $p_w < \frac{1}{nk-1}$ ✓

3. $p_w = 0$

4. $p_w = \frac{1}{(nk-1)}$

14) Sampling technique which selects every 8th bottle from a lot is called,

1. Simple random Sampling
2. Cluster Sampling
3. Systematic sampling ✓
4. Stratified Sampling.

15) For the following population $y \leftarrow c(1, 2, 3, 4, 5)$, what is the median?

- 1. 2
- 2. 15
- 3. 3 ✓
- 4. 5

16) In Systematic Sampling, when $N=18$ with $n=3$ then what will be value of k ?

- 1. 6 ✓
- 2. 8
- 3. 5
- 4. 4

17) What is the R command for selecting a sample of size 10 from the ... 0 1.

17). What is the R command for selecting a sample of size 10 from the population

$yp \leftarrow c(11, 150, 121, 198, 12, 136, 14, 129, 17, 115, 186, 110, 121, 15, 14)$

1. $S \leftarrow \text{sample}(yp, 1)$ 2. $S \leftarrow \text{sample}(yp, 10) \checkmark$
3. $S \leftarrow \text{sample}(yp, 5)$ 4. $S \leftarrow \text{sample}(yp, 25)$

18). In systematic Sampling, population is 200 and selected sample size is 50 the sample interval is:

1. 40 2. 4 $\checkmark$
3. 250 4. 200

19) - When population is following the linear trend for $n > 1$:

1. $\text{Var}(\bar{y}_{st}) = \text{Var}(\bar{y}_{sy}) = \text{Var}(\bar{y}_{wor})$
2. $\text{Var}(\bar{y}_{st}) > \text{Var}(\bar{y}_{st}) > \text{Var}(\bar{y}_{wor})$
3. $\text{Var}(\bar{y}_{st}) < \text{Var}(\bar{y}_{st}) < \text{Var}(\bar{y}_{st})$ ✓

20) - Which of the following would generally require the largest sample size.

1. Systematic Sampling
2. Cluster Sampling
3. ~~Simple~~ Simple Random Sampling ✓
4. Proportional stratified Sampling

21) When population is following the linear trend for $n \geq 1$.

1. $\text{Var}(\bar{y}_{str}) \leq \text{Var}(\bar{y}_{ssy}) \leq \text{Var}(\bar{y}_{wor}) \checkmark$
2. $\text{Var}(\bar{y}_{str}) \geq \text{Var}(\bar{y}_{ssy}) \geq \text{Var}(\bar{y}_{wor})$
3. $\text{Var}(\bar{y}_{str}) = \text{Var}(\bar{y}_{ssy}) = \text{Var}(\bar{y}_{ssy})$

22) The list of all units in population is called:

1. Bias
2. Parameter.
3. Random Sampling
4. Sampling frame. $\checkmark$

23) 18 $\frac{N}{2} \frac{M}{2} \dots$

23). If $\sum_{i=1}^N \sum_{j=1}^M y_{ij} = 1700$, $M=5$ and $n=4$,

then weighted mean is:

1. 85 ✓

2. 90

3. 80

4. 82

$$= \frac{1700}{5 \times 4}$$

24) If $\sum_{i=1}^N \sum_{j=1}^M y_{ij} = 1745$ and $MN = 15$

then the population mean is =

1. 178.36

2. 117.33

3. 116.33 ✓

4. 120.33

$$\bar{y} = \frac{\sum_{i=1}^N \sum_{j=1}^M y_{ij}}{MN} = \frac{1745}{15} = 116.33$$

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then weighted mean is:

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2. 90

3. 80

4. 82

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then the population mean is:

1. 178.36

2. 117.33

3. 116.33 ✓

4. 120.33

$$\bar{y} = \frac{\sum_{i=1}^N \sum_{j=1}^M y_{ij}}{MN} = \frac{1745}{15} = 116.33$$

25). Meom of Cluster mean is the unbiased estimator of population meom when:

1. Clusters ~~mean~~ are equal ✓
2. Unequal
3. Both equal and unequal size
4. None.

26) A sample which is free from bias is called.

1. Biased
- 2) Unbiased. ✓
3. Negatively biased
- 4). Positively biased.

27). In statistical analysis, sample size is considered large if.

27). In statistical analysis, sample size is considered large if.

1. $n < \text{or} = 50$

2.) $n < \text{or} = 30$

3. $n > \text{or} = 30 \checkmark$

4. $n > \text{or} = 50$

28). When each member of a population has an equally likely chance of being selected, this is called.

1. Random Sampling $\checkmark$

2. Stratified Sampling.

3. Systematic

4. Cluster.

29). How many samples can be drawn by

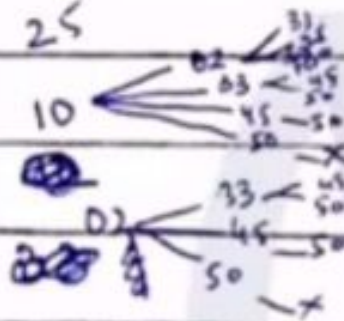
29). How many samples can be drawn by simple random sampling without replacement of size 3 from the following population 10, 02, 33, 45, 50.

1. 20

2. 10 ✓

3. 25

4. 30



45 — 50 — X

50 —>

33 < 45 — 50 — X

Mathematician

30). In sampling with Replacement a sampling unit can be selected.

1. only once. ✓
2. less than one time
3. none
4. more than one time.

31). The process of drawing a sample from a population is known as.

1. Sampling ✓
2. Census.
3. Survey sampling
4. None of these

32) what is R command for selecting the sample size $n=2$ by probability proportion to size (PPS).

32) what is R command for selecting the sample size $n=2$ by probability proportion to size (PPS).

1. $S \leftarrow \text{sample}(1:N, n=5, \text{replace} = \text{TRUE}, \text{prob} = p_i)$
2. $S \leftarrow \text{sample}(1:N, n=2, \text{replace} = \text{TRUE}, \text{prob} = p_i)$ ✓
3. $S \leftarrow \text{sample}(1:N, n, \text{replace} = \text{TRUE}, \text{prob} = p_i)$ ✗
4. $S \leftarrow \text{sample}(1:N=2n, \text{replace} = \text{TRUE}, \text{prob} = p_i)$

33) How many samples can be drawn by SRS without Replacement of size 3 from the following population 10, 02, 33, 45-

1. 4 ✓

2. 6

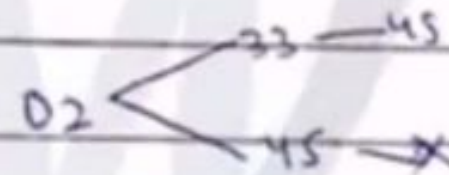
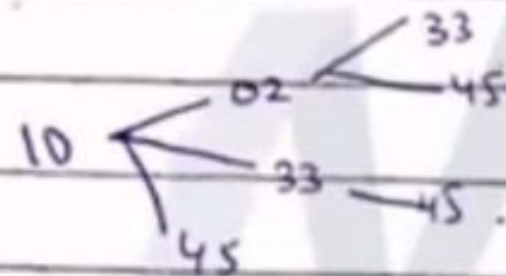
33) How many samples can be drawn by SRS without Replacement of size 3 from the following population 10, 02, 33, 45-

1.4 ✓

2. 6

3.5

4. 8



33 — x

45 →

34) Which of the following would usually require the smallest sample size because of its efficiency?

1. Quota Sampling
2. Simple random Sampling ✓
3. One-stage cluster Sampling
4. Two-stage cluster Sampling.

35) Who estimated the population of England by using auxiliary variable?

1. Grant (1962) ✓
2. Grant (1963)
3. Grant (1962)
3. Grant (1963)

36) Auxiliary variable

36) Auxiliary variable is also called

1. Supporting variable
2. Supplementary variable.
3. Both Supporting and Supplementary variable ✓
4. Independent variable

37) Which of the following methods is the best way to select a group of people for a study if you are interested in making statements about the smaller population?

1. Random Sampling
2. Quota Sampling ✓
3. Purposive Sampling
4. Convenience Sampling

38) _____ is a set of elements taken from a larger population according to certain rules.

1. Sample ✓
2. Population
3. Statistic
4. Element.

39) Which one of the following is a type of non-probability sampling?

1. Snowball Sampling
2. Quota Sampling.

39) Which one of the following is a type of non-probability sampling?

1. Snowball Sampling
2. Quota Sampling
3. Convenience Sampling
4. Systematic Sampling ✓

40) Sampling technique which selects every tenth person from a community is called.

1. Simple Random Sampling
2. Stratified Sampling
3. Systematic Sampling ✓

Probabilistic Sampling Types.

1. Simple Random Sampling
2. Stratified Sampling
3. Systematic Sampling
4. Cluster Sampling

Non-Probability Sampling Types.

1. Convenience Sampling
2. Purposive Sampling
3. Quota Sampling

41) which of the following statement are true?

1. Take another sample ✓
2. Accept the lot
3. Reject the lot
4. None of these

42) In double sampling, if the numbers of defects is in between the two cut off number C_1 and C_2 then:

- 1.
- 2.
- 3.
- 4.

43) Considering sample size, sampling distribution standard error decreases when the

3.

4.

43). Considering sample size, sampling distribution standard error decreases when the

1. Size of sample decreases 2. increases. ✓
3. margin of error decreases 4. increases.

44) If: $N = 3000$, $N_1 = 1000$, $n_1 = 150$, $N_2 = 2000$, $n_1 = 15$, $n_2 = 15$, the value of w_1 is

$$\frac{1000}{15}$$

1. 0.22

2. 0.54

3. 0.44

4. 0.33 ✓

45) If - The value of w_2 is -

Q14. The value of w_2 is

1. 0.56

2. 0.50

3. 0.67 ✓

4. 0.70

46). When every member of the accessible population has an equal chance of being selected to participate in study, the research is using

1.) purposive sampling

2.) Simple random sampling ✓

3.) convenience

4.) stratified

47) Which one of the following is the benefit of using simple random sampling?

1. Information can refuse to
2. Interviewers can choose respondent freely
3. The results are always repres
4. We can calculate the accuracy of results ^{require} ✓

48) Which of the following would usually be the largest sample because of its efficiency?

1. Two stage cluster sampling
2. One stage C.S. ✓
3. Simple random
4. Quota Sampling

10 / 13

49) People who are available, volunteer, or can be easily

49) People who are available, volunteer, or can be easily recruited are used in the sampling called.

- 1. Convenience sampling ✓
- 2. Cluster
- 3. Simple
- 4. Systematic

50) Which of the following would require the largest sample size.

- 1. Systematic sample.
- 2. Simple random sampling.
- 3. Cluster sampling ✓
- 4. Proportional stratified.

9) A technique used when selecting clusters of different

9) A technique used when selecting clusters of different sizes is called.

1. Two stage sampling.
2. Probability proportional to size ✓
3. Cluster sampling ~~also~~
- 4) One-stage sampling.

42:

42 :

5) Mrs. Trahan samples her class by selecting every third person on her class list. Which type of sampling method is this.

1. Cluster
2. Systematic ✓
3. Simple
4. Stratified.

6) Farmer Joe's apple tree farm is set up in 100 rows. He counts the number of apples produced on every

sampled.

5). Farmer Joe's apple tree farm is set up in 100 rows. He counts the number of apples produced on every 10th row to estimate the number apples produced on the whole farm. This is _____ sampling.

1. Simple
2. Cluster
3. Systematic ✓
4. Stratified

11 / 13

64). Farmer Joe's separate his apple tree farm into 10 regions.

54) Farmer Joe's separate his apple tree farm into 10 regions. He counts the number of apples produced in just one of the region and uses that estimate to predict the number of apples produced on the whole farm. This is sampling.

1. Simple

2. Cluster. ✓.

3. Stratified

4. Systematic.

(5) Mrs. Trahan samples her class by selecting all students sitting at group 1 and group 5 in her class. This sampling technique is called,

1. Cluster. ✓

2. Simple

3. Systematic

4. Stratified.

(6) Mrs. Trahan samples her class by picking 10 numbers from her hat and each number is assigned to a student. This is

1. Cluster

2. Simple ✓

(5) Mrs. Trahan samples her class by selecting all students sitting at group 1 and group 5 in her class. This sampling technique is called,

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3. Systematic

4. Stratified

(6) Mrs. Trahan samples her class by picking 10 numbers from her hat and each number is assigned to a student. This is

1. Cluster

2. Simple ✓

3. Systematic

4. Stratified.

9) Who introduced the basic concept of ranked set sampling.

1. John (1952)

3. None of these

2.

4.

58) The list of all units in a population is called.

1. Bias

3. Parameter

2. Random Sampling

4. Sampling frame. ✓

59) When one examines the

55) When one examines the

1. Bias

2. Sampling

3. Census ✓

4. Sample

60) When sample size increases, everything else remaining the same, the width of a confidence interval for a population parameter will:

60) When sample size increases, everything else remaining the same, the width of a confidence interval for a population parameter will:

1. decrease ✓
2. increase.
3. remain unchanged.
4. Sometime increase, some-time decrease.

61) Which of the following statements best describes the relationship between a parameter and a statistic?

4. Sometime increase, some-time decrease.

61). Which of the following statements best describes the relationship between a parameter and a statistic?

1. A statistic is used to estimate a parameter ✓

2.

3.

4.

13 / 13

62). The value of n_2 by proportional allocation from the following information is.

62). The value of n_2 by proportional allocation from the following information is:

$$N_1 = 580, N_2 = 140, n = 80$$

L 65.

$$n_2 = \frac{N_2 n}{N}$$

$$= \frac{140 (80)}{720}$$

$$= 15.57$$

Regression method of estimation is used to get.

- A. Small variance
- B. Large variance
- C. Unbiased variance
- D. Biased but consistent variance

ANSWER: A

A questionnaire consists of set of queries about a.

- A. Specific issue
- B. Sample
- C. Population
- D. Sample design

ANSWER: A

Regression method of estimation was introduced by

Regression method of estimation was introduced by.

- A. R A Fisher
- B. Karl Pearson
- C. Watson
- D. Cochran

ANSWER: C

In regression method of estimation β is.

- A. Independent variable
- B. Dependent variable
- C. Slope of the line
- D. Regression coefficient

ANSWER: C

Regression estimation can be made in

Regression estimation can be made in.

A. One way

B. Two way

C. Three way

D. None

ANSWER: B

The bias is negligible in.

A. Large samples

B. Small samples

C. Ratio

D. Sample mean

ANSWER: A

In regression method of estimation sample mean is an unbiased estimate of the population.

- A. Total
- B. Mean
- C. Ratio
- D. Variance

ANSWER: B

In regression method of estimation sample variance is smaller than variance of.

- A. Ratio method of estimation
- B. Stratified random sample
- C. Simple random sample
- D. All

ANSWER: C

Regression method was introduced in.

- A. 1925
- B. 1937
- C. 1940
- D. 1938

ANSWER: B

Ratio method of estimation is not applicable when regression line is passing

- A. Not through origin
- B. Through origin
- C. Through y-axis
- D. Through x-axis

ANSWER: A

Sampling techniques was written by

Sampling techniques was written by.

- A. William G Cochran
- B. Ronald A Fisher
- C. Kish
- D. Deming

ANSWER: A

Sampling techniques was published by.

- A. John Wiley and Sons
- B. McGraw Hill
- C. Chapman & Hall
- D. Macmillan

ANSWER: A

We prefer ratio method when line passes.

We prefer ratio method when line passes.

- A. Through origin
- B. Y axis
- C. X axis
- D. None

ANSWER: A

The ratio estimate of population total can be done.

- A. In a separate way
- B. In a combined way

- C. In an independent way
- D. None

ANSWER: B

Bias and relative bias are two.

- A. Similar estimates
- B. Different estimates
- C. Estimators
- D. None

ANSWER: B

In ratio method of estimation sample mean is a consistent estimate of.

- A. Population mean
- B. Population total
- C. Population proportion
- D. None

ANSWER: A

A sample of cluster is selected.

A sample of cluster is selected.

- A. Randomly
- B. Systematically
- C. Independently
- D. None

ANSWER: A

Regression model with error term, independent variable and dependent variable is called.

- A. Fixed model
- B. Exact model
- C. Random model
- D. None

ANSWER: C

Systematic sampling demands the selection of single.

- A. Sampling unit
- B. Sample design
- C. Sample frame
- D. None

ANSWER: A

The estimate of ratio in ratio method of estimation is consistent estimate of.

- A. Population ratio
- B. Population mean
- C. Population total
- D. None

ANSWER: A

Regression model with stochastic term and independent variables is called.

Regression model with stochastic term and independent variables is called.

- A. Random model
- B. Fixed model
- C. Stochastic model
- D. All

ANSWER: D

Systematic sampling is imprecise when the unit in sample is.

- A. Homogeneous
- B. Heterogeneous
- C. Large
- D. None

ANSWER: A

If sample size is large sample mean to approaches to.

If sample size is large sample mean approaches to.

- A. Population mean
- B. Population variance

- C. Normal distribution
- D. None

ANSWER: A

In regression method of estimation the auxiliary variate (X_i) is correlated with.

- A. X_i
- B. Y_i
- C. R_i
- D. X_i & Y_i

ANSWER: B

b is.

- A. An estimate of change
- B. Regression coefficient
- C. Constant
- D. Variable

ANSWER: B

In regression method of estimation b and B are.

- A. Same
- B. Constant
- C. Variable
- D. sample and population regression coefficient respectively

ANSWER: D

If $b - B = 0$. It means that.

If $b - B = 0$. It means that.

- A. MSE is zero
- B. Sampling error not ignored
- C. Sampling error ignored
- D. Variance zero

ANSWER: C

To compare ratio, regression and simple random sampling, the sample size should be.

ANSWER: B

If $\rho > 0$ variance of regression is superior than.

- A. Simple random sample
- B. Ratio estimate
- C. Stratified random sampling
- D. Cluster sampling

ANSWER: A

Simple random sampling and systematic sampling are.

- A. Probability sampling
- B. Non probability sampling
- C. Quota sampling
- D. Time sampling

ANSWER: A

In systematic sampling we divide.

In systematic sampling we divide.

- A. Population into n th parts
- B. Population into n equal of kit unit each
- C. Population
- D. Sample

ANSWER: B

In systematic sampling the first selection of unit determines the.

- A. First sample
- B. Last sample
- C. Whole sample
- D. None

ANSWER: C

Regression method of estimation was introduced to increase.

- A. Variance
- B. Error
- C. Precision
- D. Accuracy

ANSWER: C

Systematic sample is as precise as stratified random sample when one unit is drawn from.

- A. Population
- B. Each unit
- C. Each group
- D. Each stratum

ANSWER: D

Systematic sampling demands the selection of single complex.

Systematic sampling demands the selection of single complex.

- A. Sampling frame
- B. Frame
- C. Sampling design
- D. Sampling unit

ANSWER: C

The cluster may be.

- A. Equal
- B. unequal
- C. Equal or unequal
- D. Constant

ANSWER: C

Clusters are known as.

- A. PSU
- B. SSU
- C. BSU
- D. MSU

ANSWER: A

The performance of systematic sampling depends on.

- A. Sample size
- B. Population
- C. Properties of population
- D. Characteristics of sample

ANSWER: C

In cluster sampling units are found in the form of

In cluster sampling units are found in the form of.

- A. Aggregate
- B. Groups
- C. Cluster
- D. Stratum

ANSWER: B

Cluster sampling is a type of.

- A. Purposive sampling
- B. Quota sampling
- C. Probability sampling
- D. Non probability sampling

ANSWER: C

Clusters are.

- A. Similar to stratum

Clusters are.

- A. Similar to stratum
- B. Homogeneous
- C. Dissimilar to stratum
- D. Systematic sample

ANSWER: C

Regression estimation can be made in.

- A. One way
- B. Two way
- A. Three way

B. None

ANSWER: B

The bias is negligible in.

- A. Large samples
- B. Small samples
- C. Ratio
- D. Sample mean

ANSWER: B

In regression method of estimation sample total is an unbiased estimate of the population.

- A. Total
- B. Mean
- C. Ratio
- D. Variance

ANSWER: A

In regression method of estimation sample ratio is an unbiased estimate of the population.

- A. Total

In regression method of estimation sample ratio is an unbiased estimate of the population.

- A. Total
- B. Mean
- C. Ratio
- D. Variance

ANSWER: C

Cluster sampling is a.

- A. Non probability sampling
- B. Probability sampling
- C. Type of survey
- D. None

ANSWER: B

Linear regression model without stochastic term is called.

- A. Exact model

Linear regression model without stochastic term is called.

A. Exact model

- B. Dummy model
- C. Random model
- D. None

ANSWER: A

When regression line passes through the origin, sample ratio is an unbiased estimate

- A. Population ratio
- B. Population mean
- C. Population proportion
- D. All

When regression line passes through the origin, sample ratio is an unbiased estimate

- A. Population ratio
- B. Population mean
- C. Population proportion
- D. All

ANSWER: A

In systematic sampling first element is selected.

- A. Randomly
- B. Independently
- C. Systematically
- D. None

ANSWER: A

Regression model with error term is called.

- A. Stochastic model
- B. Fixed model

Regression model with error term is called.

- A. Stochastic model
- B. Fixed model
- C. Exact model
- D. None

ANSWER: A

In systematic sampling subsequently elements are selected.

- A. Systematically
- B. Randomly
- C. Independently
- D. None

ANSWER: A

The variance of optimum allocation with cost function is less than variance of.

- A. Cluster sampling
- B. Systematic sampling
- C. Simple random sampling
- D. None

ANSWER: B

When regression line goes through the origin, sample mean is an unbiased estimator of

- A. Population mean
- B. Population ratio
- C. Population proportion

ANSWER: B

When regression line goes through the origin, sample mean is an unbiased estimator of

- A. Population mean
- B. Population ratio
- C. Population proportion
- D. None

ANSWER: A

Regression model with error term and many independent variables is called.

- A. Random model
- B. Exact model
- C. Fixed model

Which of the following would usually require the smallest sample size because of its efficiency?

Select the correct option

 Reload Math Equations

- ☐ Quota sampling
- ☐ Two stage cluster sampling
- ☐ One stage cluster sampling
- ☒ Simple random sampling

Click to Save Answer & Move to Next Question

Question # 2 of 2

Time Left: 87 sec(s)

Hartley-Ross proposed the unbiased regression estimator in which year:

Select the correct option

[Reload Math Equations](#)

- ☐ None of the above
- ☐ in 1955
- ☐ in 1971
- ☒ in 1954

[Click to Save Answer & Move to Next Question](#)

2 / 32

Partly sunny



Search



ENG
UK



Among these, which sampling is based on equal probability?

Select the correct option

Reload

- ☐ Probability sampling
- ☐ Systematic sampling
- ☐ Stratified random sampling
- ☒ Simple random sampling

3 / 32

Click to Save Answer & Move to

The distribution that is formed by all possible values of a statistics is known as:

Select the correct option

Reload

- ☒ Sampling distribution
- ☐ Hypergeometric distribution
- ☐ Normal distribution
- ☐ Binomial distribution

4 / 32

Click to Save Answer & Move to

Increasing the sample size has the following effect upon the sampling error?

Select the correct option

- ☐ All of the above
- ☒ It reduces the sampling error
- ☐ It has no effect on the sampling error
- ☐ It increases the sampling error

5 / 32

Question #1 of 2

If sample size is large sample mean to approaches to population mean

Select the correct option

True



False



6 / 32

Click to Save Answer

Question # 1 of 2

Sample is regarded as a subset of?

Select the correct option

- ☐ Set
- ☒ Population
- ☐ Distribution
- ☐ Data

7 / 32

The difference between the expected value of a statistic and the value of the parameter being estimated is called α :

Select the correct option

☐	Sampling error
☐	Standard error
☒	Bias
☐	Non-sampling error

8 / 32

Cluster sampling is used:

Select the correct option

- ☐ When variance is fixed
- ☒ When population units are in group
- ☐ To reduce the cost
- ☐ To reduce the sample size

Which of the following is not a type of non-probability sampling?

Select the correct option



Stratified random sampling



Snowball sampling



Quota sampling



Convenience sampling

10 / 32

Sampling error can be reduced by:

Select the correct option

- | | |
|----------------------------------|--------------------------------|
| ☐ | Increasing the population size |
| ☐ | Non-probability sampling |
| ☒ | Increasing the sample size |
| ☐ | Decreasing the sample size |

11 / 32

The difference between a statistic and the parameter is called:

Select the correct option

- | | |
|----------------------------------|----------------|
| ☒ | Sampling error |
| ☐ | Non-random |
| ☐ | Probability |
| ☐ | Random |

12 / 32

Question # 1012

Two stage sampling design is better than the single stage sampling only if the elementary units in the same units are

Select the correct option

- | | |
|----------------------------------|-----------------------|
| ☐ | Either a or b |
| ☐ | Positively correlated |
| ☐ | Independent |
| ☒ | Negatively correlated |

13 / 32

Systematic sampling means

Select the correct option

☐

Selection of n contiguous units

☐

Selection of n largest units

☒

Selection of n units situated at equal distance

☐

Selection of n middle units in a sequence

14 / 32

Click to Save Answer & Move to Next Question

Ranked set sampling was introduced by McIntyre in the year

Select the correct option

Reload Math Equations

- ☐ 1954
- ☐ 1950
- ☐ 1956
- ☒ 1952

15 / 32

Click to Save Answer & Move to Next Question

Question # 2 of 2

Time Left: 84

Double sampling is also known as

Select the correct option

 Reload Math Equation

- ☐ Two directional sampling
- ☐ Two stage sampling
- ☐ All of above
- ☒ Two phase sampling

16 / 32

[Click to Save Answer & Move to Next Question](#)

Question # 2 of 2

Time Left: 88

In random sampling, the probability of selecting an item from the population is:

Select the correct option

[Reload Math Equations](#)

- ☒ Known
- ☐ Zero
- ☐ One
- ☐ Unknown

17 / 32

[Click to Save Answer & Move to Next Question](#)

Which of the following would generally require the largest sample size?

Select the correct option

Reload Math

- ☐ Systematic sampling
- ☐ Proportional stratified sampling
- ☒ Cluster sampling
- ☐ Simple random sampling

18 / 32

The list of all units in a population is called.

Select the correct option

Reload Math Equ

- ☐ Parameter
- ☐ Statistic
- ☒ Sampling frame
- ☐ Bias

19 / 32

Click to Save Answer & Move to Next Que

Question # 2 of 2

Time

What is the R command for writing variance-covariance matrix $\Sigma = \begin{bmatrix} 1 & .95 \\ .95 & 1 \end{bmatrix}$?

Select the correct option

Reload Mat

- ☐ `sig=matrix(c(1,0.75,0.75,1),ncol=4)`
- ☐ `sig=matrix(c(1,0.75,0.75),ncol=2)`
- ☒ `sig=matrix(c(1,0.75,0.75,1),ncol=2)`
- ☐ None of the above

20 / 32

Click to Save Answer & Move to Next

What is the R command to find out variance of x?

Select the correct option

Reload Math Equ

- ☐ Var ()
- ☐ x = variance
- ☒ Var (x)
- ☐ None of the above

21 / 32

Click to Save Answer & Move to Next Question

Question # 2 of 2

Simple random sampling and systematic sampling are

Select the correct option



probability sampling



non probability sampling



time sampling



quota sampling

22 / 32

Which of the following sampling methods is the best way to select a group of people for a study if you are interested in making statements about the larger population?

Select the correct option

[Reload Math Equations](#)

- ☒ Random sampling
- ☐ Convenience sampling
- ☐ Quota sampling
- ☐ Purposive sampling

23 / 32

[Click to Solve Answer & Model Text Question](#)

When Hartley-Ross proposed the unbiased ratio estimator:

Select the correct option



In 1954



None of the above



In 1964



In 1965

24 / 32

Click to Save Answer & Move

What is the R command to find out mean of x?

Select the correct option

- ☐ `x = mean`
- ☐ None of the above
- ☒ `mean (x)`
- ☐ `Mean ()`

25 / 32

Grant estimated the population of England in which year:

Select the correct option

- ☐ in 1962
- ☒ in 1862
- ☐ in 1952
- ☐ in 1867

The estimate of ratio in ratio method of estimation is consistent estimate of population ratio.

Select the correct option

☐	False
☒	True

27 / 32

A representative part of a larger population according to certain rules is called

Select the correct option

☐	Statistic
☐	Population
☒	Sample
☐	Element

28 / 32

Auxiliary variable is also called:

Select the correct option

- ☐ sampling Variable
- ☐ None of the above
- ☐ dependent variable
- ☒ supporting variable

29 / 32

Click to Save Answer & Move to Next Question

Q. Cluster sampling is also known as..... sampling.

Select the correct option

Reload More

- ☒ Area
- ☐ Controlled
- ☐ Judgmental
- ☐ Non-probability

30 / 32

It is recommended to use the whole population rather than a sample when the population size is of what size?

Select the correct option

- ☐ 1000 or less
- ☐ You should always use a sample
- ☐ 1500 or less
- ☐ 100 or less

31 / 32

It is recommended to use the whole population rather than a sample when the population size is of what size?

Select the correct option

☐

1000 or less

☐

You should always use a sample

☐

1500 or less

☒

100 or less

31 / 32

Click to Save Answer & Move t

Estimated value of square root of 800 is:

Select the correct option

☒ 28.28

☐ 254.95

☐ 48.48

☐ 25495

32 / 32

Click to Save Answer

If
 $\text{Var}(\hat{y}_{xy}) < \text{Var}(\hat{y}_{ars})$

:

Select the correct option

 Reload Math Equations

☐ $\rho_w = 0$

☐ $\rho_w = \frac{1}{(nk-1)}$

☐ $\rho_w > -\frac{1}{(nk-1)}$

☒ $\rho_w < -\frac{1}{(nk-1)}$

34 / 65

What is the bias of ratio estimator in two stage sampling?

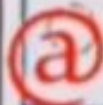
Select the correct option



$$MSE(\tilde{y}_{r2s}) \approx Y^2(V_{02} - V_{20} + 2V_{11})$$



$$MSE(\tilde{y}_{r2s}) \approx Y^2(V_{02} - V_{20} - 2V_{11})$$



$$MSE(\tilde{y}_{r2s}) \approx Y^2(V_{02} + V_{20} - 2V_{11})$$



None of the above

Question #10 of 10 (Start time: 02:07 PM, 20 July 2023)

In sampling without replacement, an element can be chosen:

Select the correct option

- | | |
|----------------------------------|-------------------|
| ☐ | More than once |
| ☐ | Less than once |
| ☒ | only once |
| ☐ | Difficult to tell |

Check

Question # 3 of 10 (Start time: 03:05:58 PM, 21 July 2023)

Total Marks

Who introduced the basic concept of ranked set sampling?

Select the correct option

 Reload Math Equations

- | | |
|-----------------------|-----------------|
| ☐ | McIntyre (1952) |
| ☐ | Muhante (1952) |
| ☐ | John (1952) |
| ☐ | Isaki (1952) |

Question # 4 of 10 (Start time: 02:06:23 PM, 21 July 2023)

Total Marks: 1

When the population size is what size, it is recommended to use the entire population instead of a sample?

Select the correct option

Reload Math Equations

- | | |
|----------------------------------|-----------------|
| ☐ | 100 or greater |
| ☒ | 100 or less |
| ☐ | 1000 or greater |
| ☐ | 1000 or less |

CamScanner

CamScanner

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STAT22 - Sampling Techniques (Quiz 4)

Quiz Start Time: 02:04 PM, 21 July 2023

Question # 5 of 10 (Start time: 02:07:04 PM, 21 July 2023)

Total Marks

Question # 6 of 10 (Start time: 02:07:50 PM, 21 July 2023)

Total Marks

If the respondents do not provide the required information to the researcher, then it is known as

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|-------------------------|
| ☐ | none of the above |
| ☐ | both of the above |
| ☒ | problem of non-response |
| ☐ | non-sampling error |

39 / 65

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 02:08:34 PM, 21 July 2023)

Total Marks: 1

When Grant estimated the population of England:

Select the correct option

Viewed Math Equations

- | | |
|---|------|
| ☒ | 1582 |
| ☐ | 1662 |
| ☐ | 1952 |
| ☐ | 1687 |

Question # 9 of 10 (Start time: 03:08:45 PM, 21 July 2023)

Problem of non-response

Select the correct option

- | | |
|-----------------------|---------------|
| ☐ | none |
| ☐ | can be solved |

41 / 65

Question # 10 of 10 (Start time: 03:10:23 PM, 21 July 2023)

Total Marks: 1

What is the R command to find out mean of x?

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---------|
| ☒ | mean(x) |
| ☐ | mean="x |

42 / 65

The term used to define the smallest element in sample selection is:

Select the correct option

- | | |
|-----------------------|------------------|
| ☐ | population unit |
| ☐ | statistical unit |
| ☐ | respondent |
| ☐ | sampling unit |
- 

Click to Show Answer & Move to Next Question

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STA632 - Sampling Techniques (Quiz 4)

Quiz Start Time: 02:51 PM, 21 July 2023

Question # 2 of 10 (Start time: 02:52:03 PM, 21 July 2023)

Total Marks:

Horvitz and Thompson introduced an unbiased estimator in:

Select the correct option

 WebCam Math Equations

- | | |
|-----------------------|------|
| ☐ | 1912 |
| ☐ | 1952 |
| ☐ | 1962 |
| ☐ | 1967 |
| ☐ | 1997 |

Question #1 of 10 (Start time: 02:51:36 PM, 21 July 2023)

Total Marks:

What is the value of combined type ratio estimator with $\bar{y}_1=18.5$, $\bar{y}_2=80.25$ and $\bar{y}=75.32$?

Select the correct option

Search Math Equations

- | | |
|----------------------------------|--------|
| ☐ | 16.323 |
| ☐ | 19.364 |
| ☒ | 17.363 |
| ☐ | 18.632 |

Click to Save Answer & Move to Next Question

45 / 65

Question # 3 of 10 (Start time: 02:44:24 PM, 21 July 2023)

Total Marks:

What is the value of combined type ratio estimator with $\bar{y}_1=15.5$, $\bar{y}_2=70.25$ and $\bar{y}=65.327$

Select the correct option

☒	14.50
☐	12.50

14.50

Submit Answer & Move to Next Question

Click to Save Answer & Move to Next Question

46 / 65

STAT2 - Sampling Techniques (Quiz 4)

Quiz Start Time: 02:41 PM, 21 July 2023

Question # 5 of 10 (Start Time: 02:47:25 PM, 21 July 2023)

Total Marks: 1

Auxiliary variable is also called:

Select the correct option

Marked Correct (1/1) Correct

- | | |
|-----------------------|----------------------|
| ☐ | dependent Variable |
| ☐ | independent Variable |
| ☐ | sampling Variable |
| ☐ | supporting variable |

supporting variable ✓

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 03:07:09 PM, 21 July 2023)

Total Marks

Which of the following sampling methods is the best way to select a group of people for a study if you are interested in making statements about the larger population?

Select the correct option

 Reload Math Equations

☐ Non-Random sampling

☒ Random sampling

48 / 65

Question #1 of 10 (Start time: 01:31:34 PM, 04 July 2023)

Which is the R command for selecting a sample of size 3 from the population `yp <- c(11,50, 121, 198, 12, 136, 14, 129, 17, 115, 186, 110, 121, 15, 14)`:

Select the correct option



`ys <- sample(yp,3)`



`ys <- sample(yp,27)`



`ys <- sample(yp,9)`

`ys <- sample(yp,12)`

Question # 2 of 10 (Start time: 01:32:18 PM, 04 July 2023)

If $N=4000$, $N_1=2000$, $n_1=150$, $N_2=1500$, $N_3=500$, $n_2=8$, $n_3=25$, the value of W_3 is:

Select the correct option

☐

0.127

☐

0.50

☐

0.23

☒

0.125

50 / 65

What is the bias of ratio estimator in two stage sampling?

Select the correct option

☐ $MSE(\bar{y}_{r,2s}) \approx \bar{Y}^2 (V_{02} - V_{20} - 2V_{11})$

☐ $MSE(\bar{y}_{r,2s}) \approx \bar{Y}^2 (V_{02} - V_{20} + 2V_{11})$

☒ $MSE(\bar{y}_{r,2s}) \approx \bar{Y}^2 (V_{02} + V_{20} - 2V_{11})$

☐ None of the above

$$MSE(\bar{y}_{r,2s}) \approx \bar{Y}^2 (V_{02} + V_{20} - 2V_{11})$$

Question # 8 of 10 (Start time: 01:37:04 PM, 04 July 2023)

If $\bar{y}_1 = 122.8$, $\bar{y}_2 = 134.6$, $\bar{y}_3 = 133.8$ and $\bar{y}_4 = 127.2$, then What will be the means of cluster means for these clusters?

Select the correct option

122.8

☐ None of the above

Question # 8 of 10 (Start timer: 01:27:04 PM, 04 July 2023)

If $\bar{y}_1 = 122.8$, $\bar{y}_2 = 134.6$, $\bar{y}_3 = 133.8$ and $\bar{y}_4 = 127.2$, then What will be the means of cluster means for three clusters?

Select the correct option

Revised Mark

☐ 127.5

☒ 128.2

☐ 132.5

☐ 126.5

52 / 65

Question # 8 of 10 (Start timer: 01:38:34 PM, 04 July 2023)

Question # 6 of 10 (Start time: 05:28:34 PM, 04 July 2023)

The below expression for variance of mean estimator under ranked set sampling is:

Select the correct option



$$V(\bar{X}^*) = \frac{1}{mk} \sum_{i=1}^k E(X_{(i)}^* - \bar{X}_{(i)}^*)^2 - \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2$$



$$V(\bar{X}^*) = \frac{1}{k^2} \sum_{i=1}^k E(X_{(i)}^* - \bar{X}_{(i)}^*)^2 - \frac{1}{k^2} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2$$



$$V(\bar{X}^*) = \frac{1}{k} \sum_{i=1}^k E(X_{(i)}^* - \bar{X}_{(i)}^*)^2 - \frac{1}{k} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2$$



$$V(\bar{X}^*) = \frac{1}{mk} \sum_{i=1}^k E(X_{(i)}^* - \bar{X}_{(i)}^*)^2 - \frac{1}{m} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2$$

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Question # 7 of 10 (Start time: 05:48:05 PM, 04 July 2023)

Total

$$V(\bar{X}) = \frac{1}{n^2} \sum_{i=1}^n E(X_{(i)}^2 - X_{(i)}^2) = \frac{1}{n} \sum_{i=1}^n (X_{(i)}^2 - \bar{X})$$

Question # 7 of 10 (Start time: 01:40:11 PM, 04 July 2023)

Total

$$\text{Var}(\hat{\beta}_{OLS}) < \text{Var}(\hat{\beta}_{GLS})$$

is

Select the correct option

Revised Mark 10



$$\rho_{\text{res}} > 0$$



$$\rho_{\text{res}} = 0$$



None of above



$$\rho_{\text{res}} < 0$$

54 / 65

Question # 8 of 10 (Start time: 01:37:04 PM, 04 July 2023)

Total

Question # 8 of 10 (Start time: 01:37:04 PM, 04 July 2023)

If $\bar{y}_1 = 122.8$, $\bar{y}_2 = 134.6$, $\bar{y}_3 = 133.8$ and $\bar{y}_4 = 127.2$, then What will be the means of cluster means for three clusters?

Select the correct option

- ☐ 127.5
- ☒ 128.2
- ☐ 132.5
- ☐ 126.5

55 / 65

of 10 (Start time: 01:41:38 PM, 04 July 2023)

Question # 8 of 10 (Start time: 01:41:38 PM, 04 July 2023)

$$\text{Var}(\hat{y}_{reg}) < \text{Var}(\hat{y}_{OLS})$$

Select the correct option

☐

$$\rho_{\text{test}} > 0$$

☐

None of above

☒

$$\rho_{\text{test}} < 0$$

☐

$$\rho_{\text{test}} = 0$$

56 / 65

Question # 8 of 10 (Start time: 01:42:04 PM, 04 July 2023)

What is the value of combined type ratio estimator with $\bar{y}_1 = 18.5$, $\bar{y}_2 = 60.25$ and $\bar{y}_3 = 75.32$?

Select the correct option

- ☐ 15.836
- ☐ 21.521
- ☐ 20.363
- ☒ 17.363

57 / 65

Question # 2 of 10 (Start time: 12:45:11 PM, 21 July 2023)

When the population size is what size, it is recommended to use the entire population instead of a sample?

Select the correct option

- ☒ 100 or less
- ☐ 1000 or greater
- ☐ 100 or greater
- ☐ 1000 or less

58 / 65 (Start time: 12:46:48 PM, 21 July 2023)

What is the R command to find out variance of x?

1000 or less

Question # 3 of 10 (Start time: 12:46:48 PM, 21 July 2023)

What is the R command to find out variance of x?

Select the correct option

☒ var(x)

☐ X = variance

59 / 65

Problem of non-response

Select the correct option

- ☐ none
- ☐ no solution
- ☐ no meaning
- ☒ can be solved

Q . Define Nonresponse?

Ans: The individuals chosen for the sample are not ready to participate in the survey. It is called nonresponse. This is a type of selection bias.

60 / 65

If the respondents do not provide the required information to the researcher, then it is known as

Select the correct option

☒ problem of non-response

☐ none of the above

☐ non-sampling error

☐ both of the above

Q . Define Nonresponse?

Ans: The individuals chosen for the sample are not ready to participate in the survey. It is called nonresponse. This is a type of selection bias.

Question # 8 of 10 (Start time: 12:53:28 PM, 21 July 2022)

Total

61 / 65

Who introduced the basic concept of ranked set sampling?

Question # 8 of 10 (Start time: 12:53:28 PM, 21 July 2023)

Who introduced the basic concept of ranked set sampling?

Page#194

Select the correct option

- | | |
|----------------------------------|-----------------|
| ☐ | Muhante (1952) |
| ☐ | Isaki (1952) |
| ☐ | John (1952) |
| ☒ | McIntyre (1952) |
- Introduction:**
Ranked set sampling (RSS) is an alternative to simple random sampling that can sometimes offer large improvements in precision. McIntyre (1952) introduced the basic concept of ranked set sampling in order to estimate the population means of pasture and forage yields. The RSS

Reload Mock

62 / 65

Click to Save Answer & Move to Next

Horvitz and Thompson introduced an unbiased estimator in:

Select the correct option



1952



1897



1967

1862

63 / 65

Q. The Horvitz Thompson Estimator?

Answer:

> HT Estimator

- The general theory of unequal probabilities sampling without replacement presented firstly by Horvitz and Thompson (1952).

Question # 6 of 10 (Start time: 01:40:24 PM, 24 July 2023)

Total

The complex exponential function e^z is onto function.

Select the correct option

True



False



64 / 65

6 of 10 (Start time: 01:29:56 PM, 24 July 2023)

Total Marks: 1

Question # 5 of 10 (Start time: 01:29:56 PM, 24 July 2023)

Total Marks: 1

For $z = x + iy$, we have $\cos z =$ _____

Select the correct option

Related Math Questions



$$\cos x \cosh y - i \sin x \sinh y$$



$$\sin x \sinh y + i \cos x \cosh y$$



$$\sin x \cosh y + i \cos x \sinh y$$



$$\cos x \cosh y + i \sin x \sinh y$$

65 / 65

Cluster sampling is used:

Select the correct option

 Reload Math Equations

☐	When variance is fixed
☐	When population units are in group
☒	To reduce the cost
☐	To reduce the sample size

Q. Cluster sampling is also known as _____ sampling.

Select the correct option

 Reload Math Equations

☒	Area
☐	Controlled
☐	Judgmental
☐	Non-probability

Sampling error can be reduced by:

Select the correct option

 Reload Math Equations

☒	Increasing the sample size
☐	Increasing the population size
☐	Non-probability sampling
☐	Decreasing the sample size

Sample is regarded as a subset of?

Select the correct option

 Reload Math Equations



Set



Data



Population



Distribution

Randomized technique was first proposed by S. L. Warner in:

elect the correct option

 Reload Math Equations

☐	1975
☒	1965
☐	1963
☐	1985

Among these, which sampling is based on equal probability?

Select the correct option

 Reload Math Equations

☐	Stratified random sampling
☐	Systematic sampling
☐	Probability sampling
☒	Simple random sampling

The distribution that is formed by all possible values of a statistics is known as:

Select the correct option

 Reload Math Equations



Hypergeometric distribution



Normal distribution



Sampling distribution



Binomial distribution

Ranked set sampling was introduced by McIntyre in the year

Select the correct option

 Reload Math Equations

☐

1950

☐

1954

☐

1956

☒

1952

The list of all units in a population is called:

Select the correct option

 Reload Math Equations

☐	Bias
☐	Parameter
☒	Sampling frame
☐	Statistic

Systematic sampling means

Select the correct option

 Reload Math Equations



Selection of n largest units



Selection of n units situated at equal distance



Selection of n middle units in a sequence



Selection of n contiguous units

In random sampling, the probability of selecting an item from the population is:

Select the correct option

 Reload Math Equations



Known



One



Unknown



Zero

Which of the following would generally require the largest sample size?

Select the correct option

 Reload Math Equations

☒	Cluster sampling
☐	Systematic sampling
☐	Proportional stratified sampling
☐	Simple random sampling

Two stage sampling design is better than the single stage sampling only if the elementary units in the same units are

Select the correct option

 Reload Math Equations

☐	Positively correlated
☐	Either a or b
☒	Negatively correlated
☐	Independent

What is the R command for writing variance-covariance matrix $\Sigma = \begin{bmatrix} 1 & .95 \\ .95 & 1 \end{bmatrix}$?

Select the correct option

 Reload Math Equations

☐

`sig=matrix(c(1,0.75,0.75),ncol=2)`

☐

None of the above

☒

`sig=matrix(c(1,0.75,0.75,1),ncol=2)`

☐

`sig=matrix(c(1,0.75,0.75,1),ncol=4)`

Double sampling is also known as

Select the correct option

 Reload Math Equations

☐	Two stage sampling
☐	Two directional sampling
☐	All of above
☒	Two phase sampling

What is the R command to find out variance of x?

Select the correct option

 Reload Math Equations

☐	none of the above
☒	var (x)
☐	var ()
☐	x = variance

What is the R command to find out mean of x?

Select the correct option

 Reload Math Equations

☒	mean (x)
☐	Mean ()
☐	None of the above
☐	x = mean

Hartley-Ross proposed the unbiased regression estimator in which year:

Select the correct option

 Reload Math Equ

- | | |
|----------------------------------|-------------------|
| ☒ | In 1954 |
| ☐ | None of the above |
| ☐ | In 1971 |
| ☐ | In 1965 |

When Hartley-Ross proposed the unbiased ratio estimator:

Select the correct option

 Reload Math Equ



None of the above



In 1965



In 1964



In 1954

It is recommended to use the whole population rather than a sample when the population size is of what size?

Select the correct option

 Reload Math Equations

☐	You should always use a sample
☒	100 or less
☐	1500 or less
☐	1000 or less

Which of the following would usually require the smallest sample size because of its efficiency?

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|----------------------------|
| ☐ | Two stage cluster sampling |
| ☐ | One stage cluster sampling |
| ☒ | Simple random sampling |
| ☐ | Quota sampling |

Auxiliary variable is also called:

Select the correct option

 Reload Math Equations



supporting variable



None of the above



dependent variable



sampling Variable

Grant estimated the population of England in which year:

select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---------|
| ☒ | In 1662 |
| ☐ | In 1667 |
| ☐ | In 1952 |
| ☐ | In 1962 |

A representative part of a larger population according to certain rules is called

Select the correct option

 Reload Math Equations



Population



Element



Statistic



Sample

Which of the following sampling methods is the best way to select a group of people for a study if you are interested in making statements about the larger population?

Select the correct option

 Reload Math Equations

☒	Random sampling
☐	Convenience sampling
☐	Quota sampling
☐	Purposive sampling

1). How many major types of Sampling?

1. 4
2. 2 ✓
3. 8
4. 3

2). Systematic Sampling is a type of

1. Probability Sampling ✓
2. Non-Probability Sampling.

3). A sample is a representative part of

1. Population ✓
2. Self-distribution

4). Quota Sampling is a type of

1. Probability
2. Non-Probability ✓

5). A numerical quantity computed from population is:

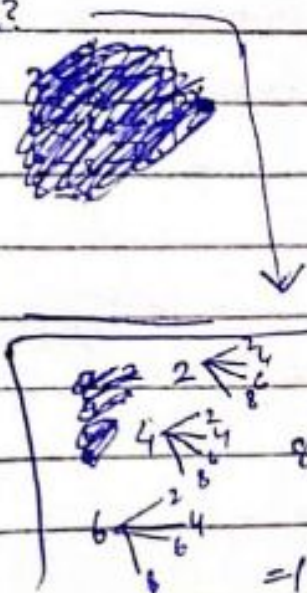
1. Parameter ✓
2. Observation
3. Statistic
4. Variable

6). If we have a population 2, 4, 6, 8 and $n=2$.
How many possible samples will be selected from sampling with replacement?

1. 12
2. 8
3. 16 ✓
4. 6

7). Cluster Sampling is a type of

1. Probability Sampling ✓
3. Non-Probability Sampling



8) In Sampling with replacement a sampling unit can be selected.

1. Only once
2. less than one
3. more than one time ✓
4. none of these.

9) Which one of these sampling is probability method

1. Quota Sampling
2. Convenience sampling
3. Cluster Sampling ✓
4. Judgment sampling

10) The other name of standard deviation of a Sampling distribution is

1. Replication Error
2. Sampling Error
3. Standard Error ✓
4. Sample Standard Deviation

11) If we have population 1, 2, 3 and $n=2$. How many samples will be selected from sampling without replacement.

1. 3 ✓
2. 9
3. 6
4. 8

$$\begin{array}{l} 1 \xrightarrow{2} 3 \quad 2 \rightarrow 3 \\ 3 \rightarrow x = 3 \end{array}$$

12) For the following population $x \sim c(1, 2, 3, 4, 5)$, what will be R command for finding mean?

1. `mean(c)`
2. `mean(x)` ✓
3. `mean x`
4. `mean{x}`

13) If $\text{Var}(\bar{y}_{sy}) < \text{Var}(\bar{y}_{src})$

1. $p_w > \frac{1}{(nk-1)}$
2. $p_w < \frac{1}{nk-1}$ ✓
3. $p_w = 0$
4. $p_w = \frac{1}{(nk-1)}$

14) Sampling technique which selects every 8th bottle from a lot is called.

1. Simple random Sampling
2. Cluster Sampling
3. Systematic sampling ✓
4. Stratified Sampling

15) For the following population $y = c(1, 2, 3, 4, 5)$, what is the median?

1. 2
2. 15
3. 3 ✓
4. 5

16) In Systematic Sampling, when $N=18$ with $n=3$ then what will be value of k ?

1. 6 ✓
2. 8
3. 5
4. 4

17) What is the R command for selecting a sample of size 10 from the population

$y_p = c(11, 150, 121, 198, 12, 136, 14, 129, 17, 115, 186, 110, 121, 15, 14)$

1. $s \leftarrow \text{sample}(y_p, 1)$
2. $s \leftarrow \text{sample}(y_p, 10)$ ✓
3. $s \leftarrow \text{sample}(y_p, 5)$
4. $s \leftarrow \text{sample}(y_p, 25)$

18) In systematic Sampling, population is 200 and selected sample size is 50 the sample interval is:

1. 40
2. 4 ✓
3. 250
4. 200

19) When population is following the linear trend for $n > 1$:

1. $\text{Var}(\bar{y}_{st}) = \text{Var}(\bar{y}_{sy}) = \text{Var}(\bar{y}_{wor})$
2. $\text{Var}(\bar{y}_{st}) > \text{Var}(\bar{y}_{st}) > \text{Var}(\bar{y}_{wor})$
3. $\text{Var}(\bar{y}_{st}) < \text{Var}(\bar{y}_{st}) < \text{Var}(\bar{y}_{st})$ ✓

20) Which of the following would generally require the largest sample size.

1. Systematic Sampling
2. Cluster Sampling
3. ~~Simple~~ Simple Random Sampling ✓
4. Proportional stratified Sampling

21) When population is following the linear trend for $n \geq 1$

1. $\text{Var}(\bar{y}_{str}) \leq \text{Var}(\bar{y}_{ssy}) \leq \text{Var}(\bar{y}_{wor})$ ✓
2. $\text{Var}(\bar{y}_{str}) \geq \text{Var}(\bar{y}_{ssy}) \geq \text{Var}(\bar{y}_{wor})$
3. $\text{Var}(\bar{y}_{str}) = \text{Var}(\bar{y}_{ssy}) = \text{Var}(\bar{y}_{ssy})$

22) The list of all units in population is called:

1. Bias
2. Parameter.
3. Random Sampling
4. Sampling frame ✓

23) If $\sum_{i=1}^N \sum_{j=1}^M y_{ij} = 1700$, $M=5$ and $n=4$,

then weighted mean is:

1. 85 ✓
2. 90
3. 80
4. 82

$$= \frac{1700}{5 \times 4}$$

24) If $\sum_{i=1}^N \sum_{j=1}^M y_{ij} = 1745$ and $MN = 15$

then the population mean is:

1. 178.36
2. 117.33
3. 116.33 ✓
4. 120.33

$$\bar{y} = \frac{\sum_{i=1}^N \sum_{j=1}^M y_{ij}}{MN} = \frac{1745}{15} = 116.33$$

25). Mean of Cluster mean is the unbiased estimator of population mean when:

1. Clusters ~~mean~~ are equal ✓
2. Unequal
3. Both equal and unequal size
4. None.

26) A sample which is free from bias is called.

1. Biased
2. Unbiased. ✓
3. Negatively biased
4. Positively biased.

27). In statistical analysis, sample size is considered large if.

1. $n < \text{or} = 50$
2. $n < \text{or} = 30$
3. $n > \text{or} = 30$ ✓
4. $n > \text{or} = 50$

28). When each member of a population has an equally likely chance of being selected, this is called.

1. Random Sampling ✓
2. Stratified Sampling.
3. Systematic
4. Cluster.

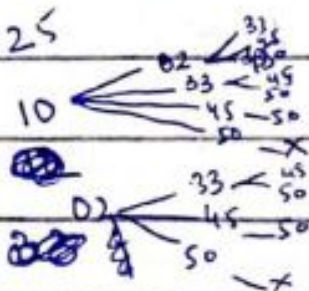
29). How many samples can be drawn by simple random sampling without replacement of size 3 from the following population 10, 02, 33, 45, 50.

1. 20

2. 10 ✓

3. 25

4. 30



45 — 50 — X

50 —>

33 — 45 — 50 — X

30) In Sampling without Replacement a sampling unit can be selected.

1. only once. ✓
2. less than one time
3. none
4. more than one time.

31) The process of drawing a sample from a population is known as.

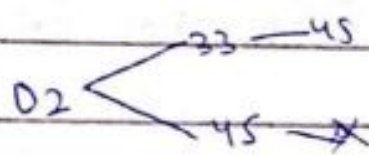
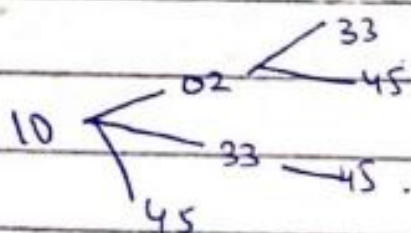
1. Sampling ✓
2. Census.
3. Survey sampling
4. None of these

32) What is R command for selecting the sample size $n=2$ by probability proportion to size (PPS).

1. `S <- sample(1:N, n=5, replace=TRUE, prob=pi)`
2. `S <- sample(1:N, n=2, replace=TRUE, prob=pi)` ✓
3. `S <- sample(1:N, n, replace=TRUE, prob=pi)`
4. `S <- sample(1:N=2, n, replace=TRUE, prob=pi)`

33) How many samples can be drawn by SRS without Replacement of size 3 from the following population 10, 02, 33, 45-

1. 4 ✓
2. 6
3. 5
4. 8



33 - x

45 - x

34) Which of the following would usually require the smallest sample size because of its efficiency?

1. Quota Sampling
2. Simple random Sampling ✓
3. One-stage cluster Sampling
4. Two-stage cluster Sampling.

35) Who estimated the population of England by using auxiliary variable:-

1. Grant (1962) ✓
2. Grant (1963)
3. Grant (1962)
3. Grant (1963)

36) Auxiliary variable is also called

1. Supporting variable
2. Supplementary variable.
3. Both Supporting and Supplementary variable ✓
4. Independent variable

37) Which of the following methods is the best way to select a group of people for a study if you are interested in making statements about the smaller population?

1. Random Sampling
2. Quota Sampling ✓
3. Purposive Sampling
4. Convenience Sampling.

38) _____ is a set of elements taken from a larger population according to certain rules.

1. Sample ✓
2. Population
3. Statistic
4. Element.

39) Which one of the following is a type of non-probability sampling?

1. Snowball Sampling
2. Quota Sampling
3. Convenience Sampling
4. Systematic Sampling ✓

40) Sampling technique which selects every tenth person from a community is called.

1. Simple Random Sampling
2. Stratified Sampling
3. Systematic Sampling ✓
4. Cluster Sampling

Probabilistic Sampling Types.

1. Simple Random Sampling
2. Stratified Sampling
3. Systematic Sampling
4. Cluster Sampling

Non-Probability Sampling Types.

1. Convenience Sampling
2. Purposive Sampling
3. Quota Sampling
4. Snow-ball Sampling

41) Which of the following statements are true?

1. Take another sample ✓
2. Accept the lot
3. Reject the lot
4. None of these

42) In double sampling, if the numbers of defects is in between the two cut off number C_1 and C_2 then:

- 1.
- 2.
- 3.
- 4.

43) Considering sample size, sampling distribution standard error decreases when the

1. Size of sample decreases
2. increases ✓
3. margin of error decreases
4. increases

Q15: $N = 3000$, $N_1 = 1000$, $n_1 = 150$, $N_2 = 2000$, $n_1 = 15$, $n_2 = 15$, the value of w_1 is

$$\frac{1000}{15}$$

1. 0.22
2. 0.54
3. 0.44
4. 0.33 ✓

Q16. The value of w_2 is

1. 0.56
2. 0.50
3. 0.67 ✓
4. 0.70

46) When every member of the accessible population has an equal chance of being selected to participate in study, the research is using

- 1) purposive sampling
- 2) Simple random sampling ✓
- 3) convenience
- 4) stratified

47) Which one of the following is the benefit of using simple random sampling?

1. Information can refuse to -
2. Interviewer can choose respondent freely
3. The results are always represent
4. We can calculate the accuracy of results ^{require} ✓

48) Which of the following would usually require the largest sample because of its efficiency?

1. Two stage cluster sampling
2. One stage C.S. ✓
3. Simple random
4. Quota Sampling

49) People who are available, volunteer, or can be easily recruited are used in the sampling called.

1. Convenience sampling ✓
2. Cluster
3. Simple
4. Systematic

50) Which of the following would require the largest sample size.

1. Systematic sample.
2. Simple random sampling.
3. Cluster sampling ✓
4. Proportional stratified.

51) A technique used when selecting clusters of different sizes is called.

1. Two stage sampling.
2. Probability proportional to size ✓
3. Cluster sampling ~~are~~
4. One-stage sampling.

42:

5) Mrs. Trahan samples her class by selecting every third person on her class list. Which type of sampling method is this.

1. Cluster
2. Systematic ✓
3. Simple
4. Stratified.

5) Farmer Joe's apple tree farm is set up in 100 rows. He counts the number of apples produced on every 10th row to estimate the number apples produced on the whole farm. This is _____ sampling.

1. Simple
2. Cluster
3. Systematic ✓
4. Stratified

5) Farmer Joe's separate his apple tree farm into 10 regions. He counts the number of apples produced in just one of the region and uses that estimate to predict the number of apples produced on the whole farm. This is _____ sampling.

- | | |
|---------------|---------------|
| 1. Simple | 2. Cluster. ✓ |
| 3. Stratified | 4. Systematic |

5) Mrs. Trahan samples her class by selecting all students sitting at group 1 and group 5 in her class. This sampling technique is called,

1. Cluster. ✓

2. Simple

3. Systematic

4. Stratified.

56) Mrs. Trahan samples her class by picking 10 numbers from her hat and each number is assigned to a student. This is

1. Cluster

2. Simple ✓

3. Systematic

4. Stratified.

9) Who introduced the basic concept of ranked set sampling.

1. John (1952)

3. None of these

2.

4.

58) The list of all units in a population is called.

1. Bias

3. Parameter

2. Random Sampling

4. Sampling frame. ✓

59) When one examines the

1. Bias

2. Sampling

3. Census ✓

4. Sample

60) When sample size increases, everything else remaining the same, the width of a confidence interval for a population parameter will:

1. decrease ✓
2. increase.
3. remain unchanged.
4. Sometime increase, some-time decrease.

61) Which of the following statements best describes the relationship between a parameter and a statistic?

1. A statistic is used to estimate a parameter ✓
- 2.
- 3.
- 4.

62) The value of n_2 by proportional allocation from the following information is.

$$N_1 = 580, N_2 = 140, n = 80$$

L 65.

$$n_2 = \frac{N_2 n}{N}$$

$$= \frac{140 (80)}{720}$$

$$= 15.57$$

Remember me in your prayers ❤️